AER 1515 – Perception for Robotics

Lecture 2 – Probability and Information

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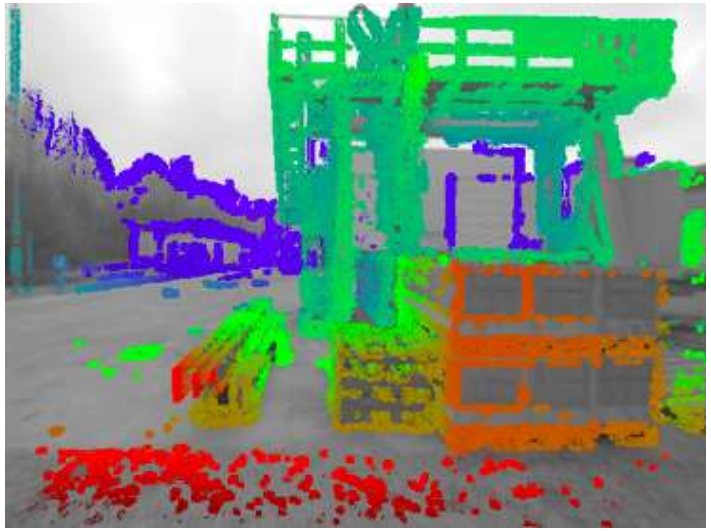


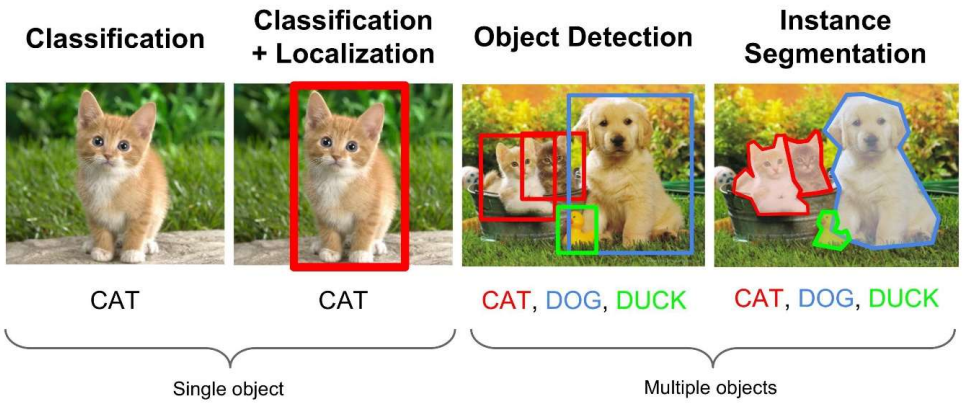
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Probability and Information Improvements

- To be added
  - Covariance and correlation (Murphy)
  - Chain rule of conditional probabilities (Bengio)
  - Beta and Dirichlet (Murphy)
  - Sample mean and variance (Murphy)
  - Mutual Information Robotics or ML Example (Murphy, others?)
  - Convexity and Jensen's inequality to allow the application of log to everything
  - Generating samples from Gaussians (or from CDFs)
  - Marginalization
  - Maholonobis distance (eigenvect/vals) (Murphy)
  - Mixture of Gaussians

Why Probability, Statistics, Information Theory





Discrete Probability Distributions

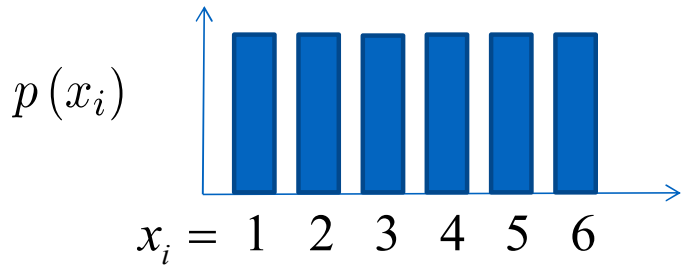
- Discrete Random Variable
  - *X* denotes a random variable
  - *X* can take on a countable number of values

$$X \in \{x_1, \ldots, x_n\}$$

  - The probability that *X* takes on a specific value must fall between 0 and 1

$$0 \leq p\left(X = x_i\right) = p\left(x_i\right) \leq 1$$

  - A 6-sided die's probability distribution – **probability mass function** (PMF)

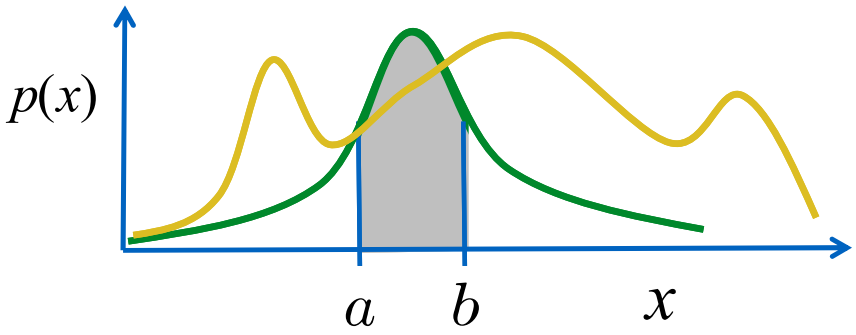


$$\sum_{i=1}^{n} p\left(x_i\right) = 1$$

Continuous Probability Distributions

- Continuous Random Variable
  - X takes on a value in a continuum
  - Probability density function

$$p(X = x) \text{ or } p(x)$$



  - Evaluated over finite intervals of the continuum

$$p(x \in (a, b)) = \int_a^b p(x)dx$$

- Note that probability is an area, probability density is the value of the pdf at a single point. A single point in a continuous distribution has zero probability of occurring.

Cumulative Distributions and Quantiles

- A **cumulative distribution function** (CDF), $P(q)$, is the probability that a random variable is less than or equal to q
  - Can be computed from the pdf

$$P(q) = \int_0^q p(x)dx$$

  - $P(q)$ is monotonically increasing because $p(x) \geq 0$, therefore $P^{-1}(\alpha)$ exists
  - A **quantile** is then a segmentation of the CDF which can be computed from its inverse
    - The median is $P^{-1}(0.5)$, upper and lower quartiles are $P^{-1}(0.75)$ and $P^{-1}(0.25)$

Measures of Distributions

- First Raw Moment
  - Mean
    - Expected value of a random variable

$$\mu = E[X]$$

$$\mu = \sum_{i=1}^{n} x_i p\left(x_i\right)$$

$$\mu = \int x p(x) dx$$

- Second Central Moment
  - Variance
    - Measure of the variability of a random variable

$$\mathrm{Var}(X) = E\left[(X-\mu)^2\right]$$

$$\mathrm{Var}(X) = \sum_{i=1}^{n} \left(x_i - \mu\right)^2 p\left(x_i\right)$$

$$\mathrm{Var}(X) = \int (x-\mu)^2 p(x) dx$$

    - square root of variance is standard deviation,

$$\sigma^2 = \mathrm{Var}(X)$$

Joint Probability and Independence

- Joint Probability
  - Probability of x and y:

$$p(X = x \text{ and } Y = y) = p(x, y)$$

    - e.g. probability of clouds and rain today

- Independence
  - If X, Y are independent, then

$$p(x, y) = p(x)p(y)$$

    - e.g. probability of two heads coin-flips in a row is ¼

Conditional Probability

- Probability of x given y

$$p(X = x \mid Y = y) = p(x \mid y)$$

  - Probability of KD for dinner, given an undergraduate Steve is cooking

- Relation to joint probability

$$p(x \mid y) = \frac{p(x, y)}{p(y)}$$

  - If X and Y are independent,

$$p(x \mid y) = p(x)$$

    - Follows from the above

Law of Total Probability

Discrete

$$\sum_x p(x) = 1$$

$$p(x) = \sum_y p(x, y)$$

$$p(x) = \sum_y p(x \mid y)\, p(y)$$

Continuous

$$\int p(x)dx = 1$$

$$p(x) = \int p(x, y)dy$$

$$p(x) = \int p(x \mid y)\, p(y)dy$$

Bayes Theorem

- From definition of conditional probability

$$p(x \mid y) = \frac{p(x,y)}{p(y)}, \quad p(y \mid x) = \frac{p(x,y)}{p(x)}$$

$$p(x \mid y)p(y) = p(x,y) = p(y \mid x)p(x)$$

- Bayes Theorem defines how to update one's beliefs about X given a known (new) value of y

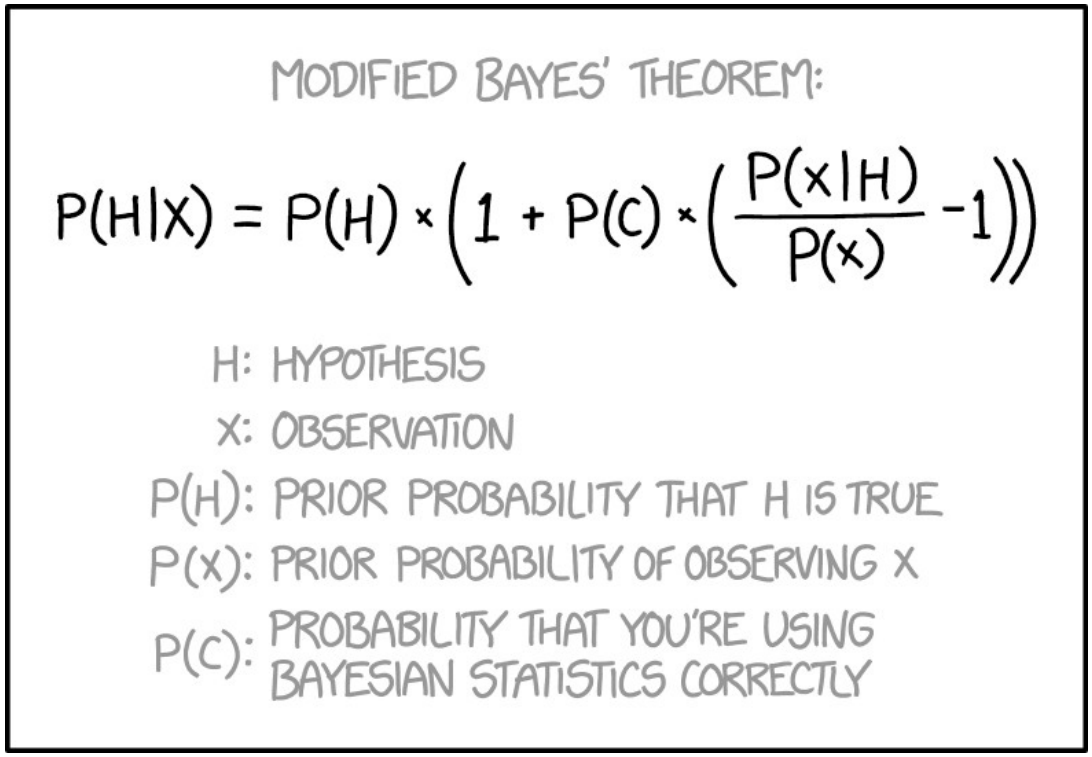
$$p(x \mid y) = \frac{p(y \mid x)p(x)}{p(y)} = \frac{likelihood \cdot prior}{evidence}$$

Probability

- Bayes Theorem
  - If Y is a measurement and X is the current vehicle state, Bayes Theorem can be used to update the state estimate given a new measurement
    - Prior: probabilities that the vehicle is in any of the possible states
    - Likelihood: probability of getting the measurement that occurred given every possible state is the true state
    - Evidence: probability of getting the specific measurement recorded

$$p(x \mid y) = \frac{p(y \mid x)\, p(x)}{p(y)} = \frac{likelihood \cdot prior}{evidence}$$

Important Qualification



Common Distributions

- **Uniform**: Defined as equal probability mass or constant probability density

$$p(x) = 1/n \qquad\qquad p(x) = \frac{1}{b-a}$$

  - Often used as an uninformative prior
  - Maximizes entropy

- **Bernoulli**
  - Let $X \sim \text{Ber}(\theta)$ be a binary random variable, then X has a Bernoulli distribution with probability of success, θ

$$X \in \{0, 1\}$$

  - The Bernoulli distribution is then $\text{Ber}(x|\theta) = \begin{cases} \theta & \text{if } x = 1 \\ 1-\theta & \text{if } x = 0 \end{cases}$

Common Distributions

- Binomial Distribution
  - Given n realizations of a binary random variable, the number of positive values is a random variable $X \sim \text{Bin}(n, \theta)$ drawn from $X \in \{0, \ldots, n\}$ where

$$\text{Bin}(k|n, \theta) = \left(\begin{array}{c} n \\ k \end{array} \right) \theta^k (1-\theta)^{n-k} \qquad \left(\begin{array}{c} n \\ k \end{array} \right) = \frac{n!}{(n-k)!k!}$$

  - The Binomial distribution has the following mean and variance, assuming 0, 1 labels are meaningful

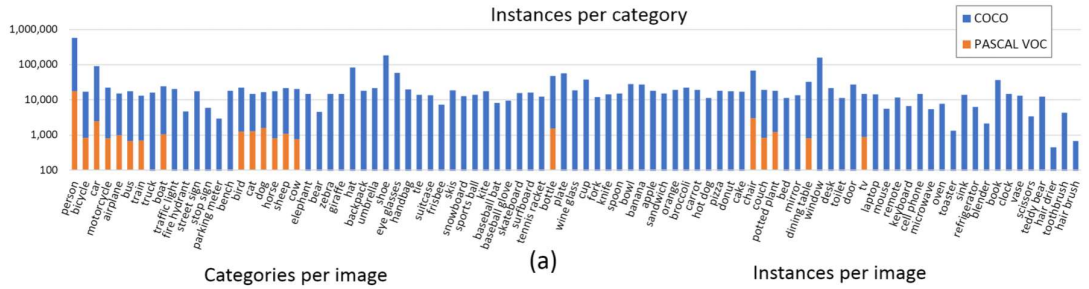
$$\mu = n\theta \qquad \text{Var} = n\theta(1-\theta)$$

Common Distributions

- **Categorical** (Discrete, Multinoulli) and **Multinomial**
  - Extension to discrete random variables that can take on more than two values
  - Categorical distribution is simply probability mass function for

$$X \in \{0, \ldots, n\}$$

$$\text{Cat}(x_i|\theta_i) = \theta_i \quad \text{if } X = x_i$$



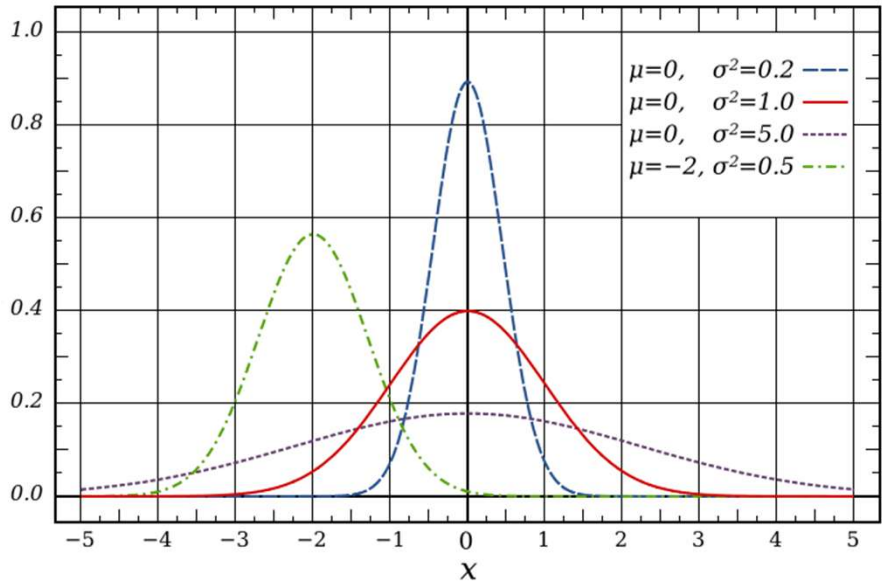
  - Multinomial distribution is analogously the probability of x_j occurrences of each of the j outcomes

$$\text{Mu}(x|n, \theta) = \begin{pmatrix} n \\ x_1 \ldots x_n \end{pmatrix} \prod_{j=1}^{K} \theta_j^{x_j}$$

Common Distributions

- Gaussian Distribution (Normal)

$$p(x) = \frac{1}{\sqrt{2\pi\sigma}} e^{\frac{-(x-\mu)^2}{2\sigma^2}} \qquad p(x) \sim N(\mu, \sigma^2)$$



Multi-variate distributions

- Vector of random variables

$$X = \begin{bmatrix} X_1 \\ \vdots \\ X_n \end{bmatrix}$$

  - Mean

$$\mu = \begin{bmatrix} \mu_1 \\ \vdots \\ \mu_n \end{bmatrix} = \begin{bmatrix} E[X_1] \\ \vdots \\ E[X_n] \end{bmatrix}$$

Covariance

- Multi-variable distributions
  - Covariance
    - Measure of how much two random variables change together

$$Cov(X_i, X_j) = E[(X_i - \mu_i)(X_j - \mu_j)]$$
$$= E[X_i X_j] - \mu_i \mu_j$$

- If $\mathrm{Cov}(X,Y)>0$, when X is above its expected value, then Y tends to be above its expected value
- If $\mathrm{Cov}(X,Y)<0$, when X is above its expected value, then Y tends to be below its expected value
- If X, Y are independent, $\mathrm{Cov}(X,Y) = 0$

Probability

- Multi-variable distribution
  - Covariance Matrix, $\sum$
    - Defines variational relationship between each pair of random variables

$$\Sigma_{i,j} = Cov(X_i, X_j)$$

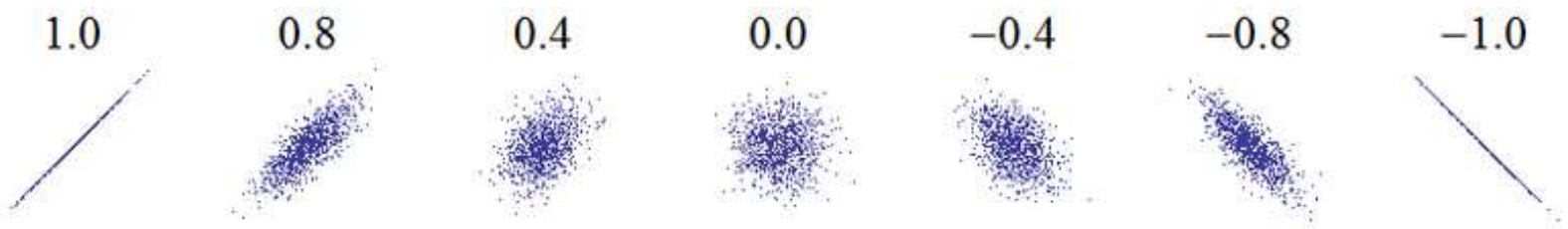
    - Generalization of variance, diagonal elements represent variance of each random variable

$$Cov(X_i, X_i) = Var(X_i)$$

    - Covariance matrix is symmetric, positive semi-definite

Covariance

- Two dimensional distributions off diagonal element covariance $Cov(X_1, X_2)$



Probability

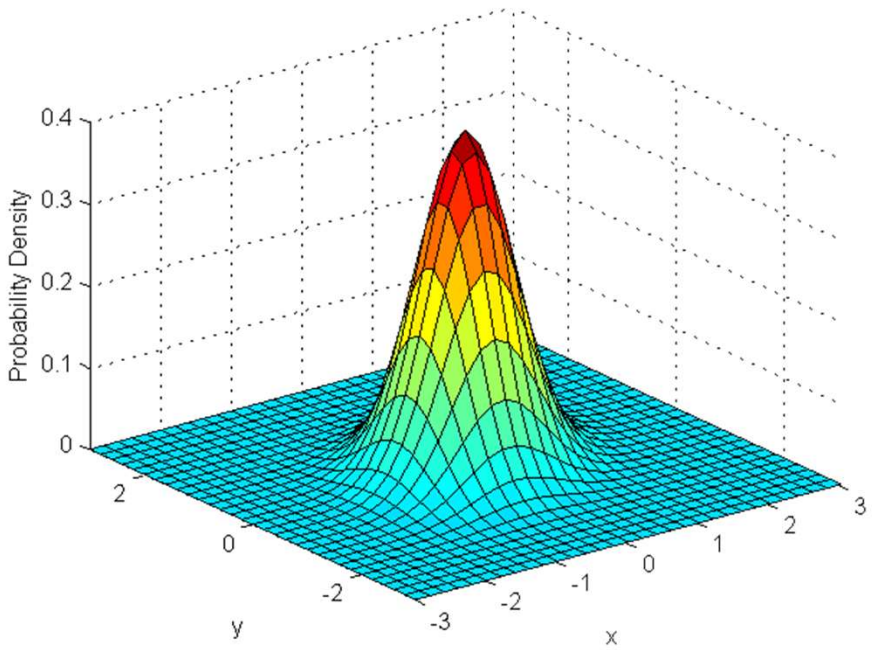
- Multiplication by a constant matrix yields

$$\begin{aligned}\operatorname{cov}(Ax) &= E\left[(Ax - A\mu)(Ax - A\mu)^T\right] \\ &= E\left[A(x-\mu)(x-\mu)^T A^T\right] \\ &= AE\left[(x-\mu)(x-\mu)^T\right] A^T \\ &= A\Sigma A^T\end{aligned}$$

Common Distributions

- Multivariate Gaussian Distribution (Normal)

$$p(\boldsymbol{x}) = \frac{1}{\sqrt{\det(2\pi\Sigma)}} e^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^T \Sigma^{-1}(\boldsymbol{x}-\boldsymbol{\mu})} \qquad p(\boldsymbol{x}) \sim N(\boldsymbol{\mu}, \Sigma)$$



Probability

- Properties of Gaussians
  - Linear combinations remain Gaussian

$$x \sim N(\mu, \Sigma), \quad y = Ax + B$$
$$y \sim N\left(A\mu + B, A\Sigma A^T\right)$$

  - Direct Product of *K* Gaussians, $N(\mu_k, \Sigma_k)$

$$p(x|\mu, \Sigma) = \prod_{k=1}^{K} p(x|\mu_k, \Sigma_k)$$

    - where

$$\Sigma^{-1} = \sum_{k=1}^{K} \Sigma_k^{-1} \qquad \mu_k = \Sigma \sum_{k=1}^{K} \Sigma_k^{-1} \mu_k$$

Gaussians are an Exponential Family

- Exponential families have many properties that make them extremely useful for statistical analysis. It can be shown that, except in a few exceptional cases, only exponential families have these properties. Examples:
  - Exponential families have sufficient statistics that can summarize arbitrary amounts of independent identically distributed data using a fixed number of values.
    - Mean, covariance are all that is needed to specify a Gaussian
  - Exponential families have conjugate priors, an important property in Bayesian statistics.
    - The posterior distribution will be of the same distribution family as the prior if the prior is a conjugate prior of the likelihood.

$$p(x \mid y) = \frac{p(y \mid x)\, p(x)}{p(y)} = \frac{likelihood \cdot prior}{evidence}$$

    - A Gaussian is a self-conjugate, meaning that if the likelihood and prior are Gaussian, then the posterior will be Gaussian as well
    - The Dirichlet distribution is a conjugate prior to the categorical distribution, meaning that samples of events can be counted and used as a prior for a categorical likelihood distribution

Generating Samples

- Generating multivariate random noise samples from Gaussian distributions
  - Define two distributions, the one of interest and the standard normal distribution

$$\delta \sim N(\mu, \Sigma) \qquad \omega \sim N(0, I)$$

  - If the covariance is full rank, it can be diagonalized
    - Symmetry implies positive semi-definiteness

$$\begin{aligned}\Sigma &= E\lambda E^T \\ &= E\lambda^{1/2} I \lambda^{1/2} E^T \\ &= HIH^T\end{aligned}$$

  - Can now relate the two distributions (linear identity)

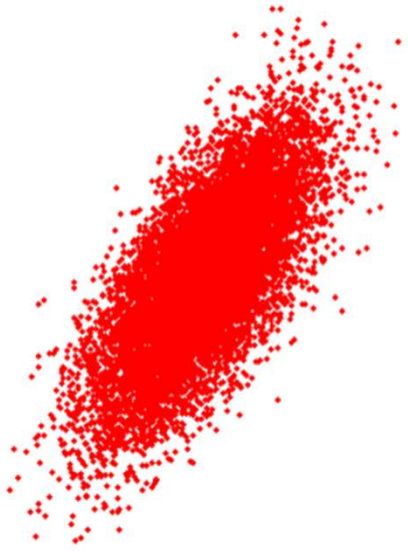
$$\delta \sim N(\mu, HIH^T)$$
$$\delta = \mu + H\omega$$

Generating Samples in Matlab/python

- To implement this in Matlab or Python for simulation purposes
  - Define $\mu, \sum$
  - Find eigenvalues , λ, and eigenvectors, E of $\sum$
  - The noise can then be created with

$$\delta = \mu + E\lambda^{1/2}\text{randn(n,1)}$$

$$\Sigma = \begin{bmatrix} 4 & 4 \\ 4 & 8 \end{bmatrix}$$



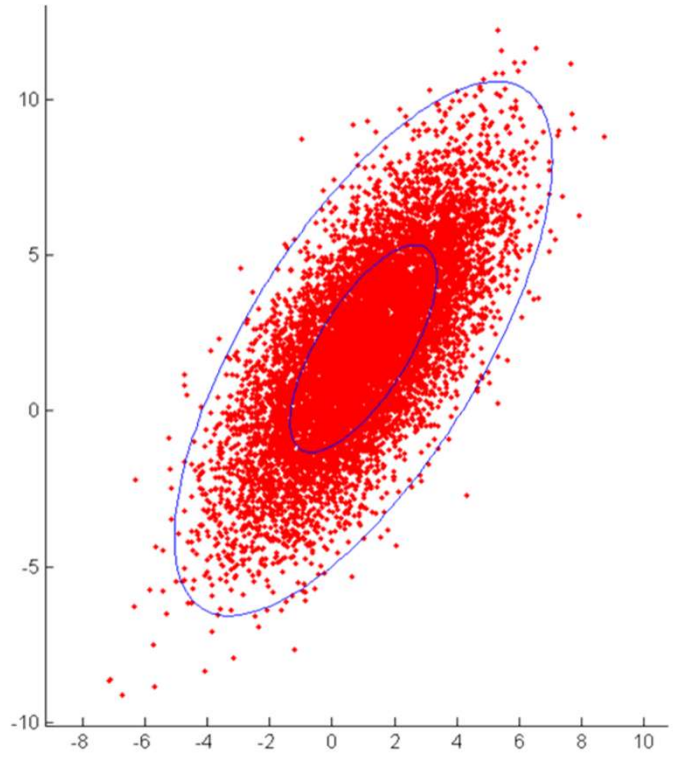
Note: numpy.random.randn in python

Probability

- Confidence ellipses
  - Lines of constant probability
    - Found by setting pdf exponent to a constant
    - Centered at the mean
    - Principal axes are eigenvectors of covariance
    - Magnitudes depend on eigenvalues of covariance

$$\mu = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} 4 & 4 \\ 4 & 8 \end{bmatrix}$$

50%, 99% error ellipses



Information Theory

- The roots of modern digital communication stem from the ground-breaking paper “A Mathematical Theory of Communication” by Claude Elwood Shannon in 1948.
  - “The fundamental problem of communication is that of reproducing at one point either exactly or approximately a message selected at another point. … Frequently the messages have meaning ... [which is] irrelevant to the engineering problem.”



- Using the following three axioms, Shannon showed that there is only one way to measure his notion of information, $I(x)$:
  - Likely events should have less information than unlikely events.
  - Guaranteed events have no information (information ≥ 0 for all events)
  - Independent events should have additive information

Information Theory

- Self-Information for a single event $X = x$

$$I(X) = -\log_* p(x) = \log_* \frac{1}{p(x)}$$

  - The base of the log simply scales the information quantity
    - * = 2 → bits
    - * = e → nats

- Shannon entropy is then the expected information in a distribution

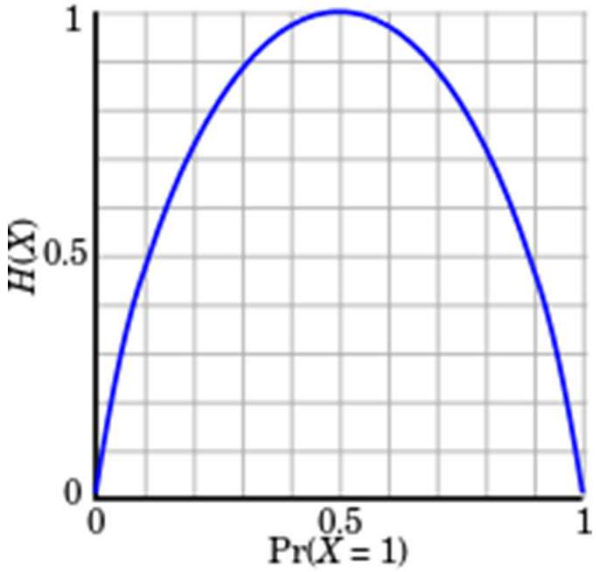
$$H(X) = E_{X \sim P}[I(x)] = -\sum_x p(x) \log_2 p(x)$$

  - Derived for the discrete case of communication of digital information through a channel
  - Therefore naturally expressed as bits with log base 2

Information Theory

- Communicating the results of repeated tossing of a die:
  - Outcomes are 1,2,3,4,5,6
  - Each occurs with equal probability 1/6
  - Entropy of distribution is (expected information)

$$
\begin{aligned}
H &= -\sum_{x=1}^{6} p(x) \log_2 p(x) \\
&= -\sum_{x=1}^{6} \frac{1}{6} \log_2 \frac{1}{6} \\
&= \log_2 6 \\
&= 2.585 \text{ bits}
\end{aligned}
$$

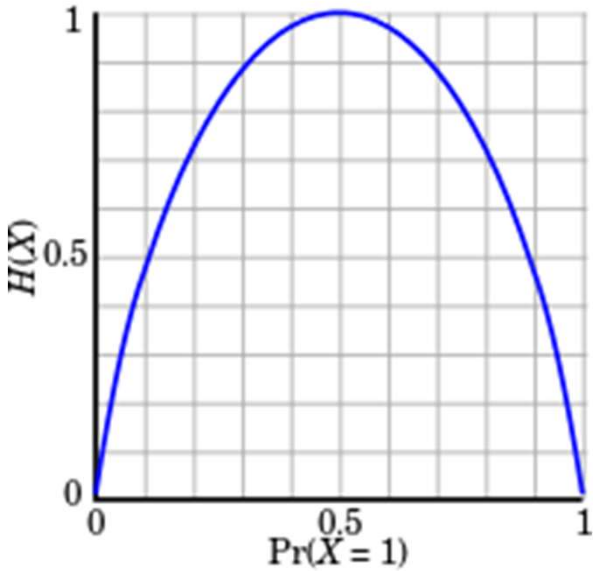


- On average, 2.585 bits needed to communicate outcome of a die toss
- In fact, the uniform distribution maximizes entropy, requires the highest number of bits

Information Theory

- Tossing a die:
  - Outcomes are 1,2,3,4,4,4
  - Probabilities 1/6, 1/6, 1/6, 1/2
  - Information provided by tossing this die is:

$$
\begin{aligned}
H &= -\sum_{x=1}^{4} p(x) \log_2 p(x) \\
&= -\left(\frac{1}{2} \log_2 \frac{1}{2} + 3 * \frac{1}{6} \log_2 \frac{1}{6} \right) \\
&= 1.792 \text{ bits}
\end{aligned}
$$



  - Leads to optimal encodings, maximum compression ratios, which can be determined based on probabilities of errors in the communication channel

Interpretation of Entropy

- At least three different ways of interpreting the term entropy
  - Communication Theory: as a measure of information
    - To reliably communicate or store the information generated by some random source X, you need no more/less than, on the average, $H(X)$ bits for each outcome
  - Thermodynamics: as a measure of disorder
    - The second law of Thermodynamics: the total entropy of an isolated system can never decrease over time.
  - Statistics: as a measure of uncertainty
    - The expectation of a loss event occurring, often referred to as risk. We'll expand on this interpretation going forward
    - Distributions with low entropy have nearly certain outcomes, distributions with high entropy are far less predictable

Information Theory

- Differential Entropy: For continuous distributions, the analog to entropy, usually expressed in nats

$$H(x) = -E[\ln(p(x))] = -\int_{-\infty}^{\infty} p(x) \ln\left(p(x)\right) dx$$

- The natural logarithm lends itself well to computations with Gaussian and other exponential distributions

- Maximum Differential Entropy Distributions
  - $[a,b]$
    - Uniform

$$H(x) = \ln(b-a)$$

  - [-∞, ∞]
    - Gaussian

$$H(x) = \frac{1}{2} \ln\left((2\pi e)^N \det \Sigma\right)$$

  - $[0, \infty]$
    - Exponential

$$H(x) = \text{exercise}$$

Kullbach-Liebler Divergence

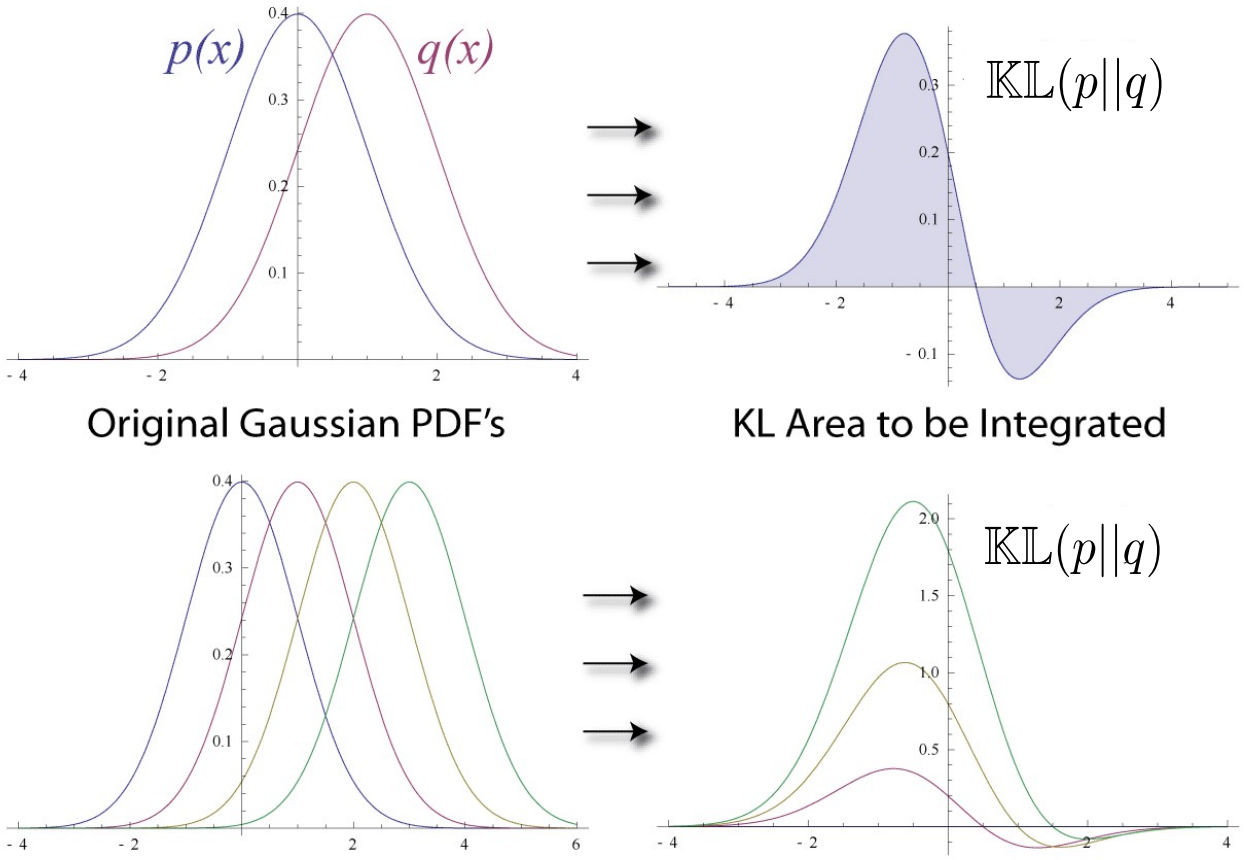
- A way to measure the dissimilarity between two distributions, *p(x)* and *q(x)* (not symmetric, so not a true distance)

$$\mathbb{KL}(p||q) = \sum_{k=1}^{K} p_k \ln \frac{p_k}{q_k}$$

- Analogously in the continuous domain

$$\mathbb{KL}(p||q) = \int_{-\infty}^{\infty} p(x) \ln \frac{p(x)}{q(x)} dx$$

KL Divergence



KL-Divergence

- The Information Inequality theorem states that for any distributions p and q:

$$\mathbb{KL}(p||q) \geq 0 \text{ and } \mathbb{KL}(p||q) = 0 \text{ iff } p(x) = q(x)$$

  - This theorem gives us our foundation for using KL-divergence as a measure of distribution similarity

  - In Bayesian inference, if q is the prior and p is the posterior, can be seen as the information gain from q to p

  - Also referred to as relative entropy, indicating its close connection to differential entropy

Cross-entropy and KL-divergence

- When there are two distributions p and q for a random variable X, cross entropy becomes useful

$$H(p,q) = \int p(x) \ln\left(\frac{1}{q(x)}\right) dx = E_p\left[\ln\left(\frac{1}{q}\right)\right]$$

- Rewriting cross entropy, we get

$$\begin{aligned} H(p,q) &= \int p(x) \ln\left(\frac{p(x)}{q(x)}\right) + p(x) \ln\left(\frac{1}{p(x)}\right) dx \\ &= \mathbb{KL}(p||q) + H(p) \end{aligned}$$

  - So cross-entropy can be thought of as the risk of using q as an approximation to p plus the uncertainty associated with p itself
  - Or as the entropy of using an encoding designed for q to encode a message drawn from p

Approximating distributions

- As hinted at on the previous slide, a distribution q can be used to approximate a distribution p
  - Let $q(\theta)$ be a distribution parametrized by θ (e.g. a Gaussian, where $\theta = (\mu,\Sigma)$, or a network)
  - Let p be the unknown distribution we'd like to approximate and from which we can draw i.i.d. samples (measurements, or training data)
  - We can minimize the KL-divergence between p and q as follows

$$\min_{\theta} \qquad \mathbb{KL}(p||q(\theta)) = H(p, q(\theta)) - H(p)$$

  - Since $H(p)$ does not depend on θ, we can equivalently minimize the cross-entropy directly

$$\min_{\theta} \qquad H(p, q(\theta)) = E_p\left[\ln\left(\frac{1}{q(\theta)}\right)\right]$$

  - This expectation can be approximated using i.i.d. samples to define a sample mean, which we'll see going forward in machine learning

$$E_p\left[\ln\left(\frac{1}{q(\theta)}\right)\right] \approx \frac{1}{n}\sum_{i=1}^{n} \ln\left(\frac{1}{q(x_i, \theta)}\right)$$

Joint Entropy and Mutual Information

- We can build on the idea of KL-divergence to explain the connection between two random variables
- If we measure the KL-divergence between the joint distribution $p(x,y)$ and the independent distribution $p(x)p(y)$, we get

$$\mathbb{KL}(p(X,Y)||p(X)p(Y)) = \int\int p(x,y)\ln\frac{p(x,y)}{p(x)p(y)}dydx$$

- This is known as the mutual information, and is a more useful measure than correlation where distributions are not simple and unimodal

$$\mathbb{I}(X;Y) = \mathbb{KL}(p(X,Y)||p(X)p(Y)) = H(X) - H(X|Y) = H(Y) - H(Y|X)$$

  - where $H(X|Y)$ is the conditional entropy, extrapolating the definition of a conditional distribution

- Thus, the Mutual Information is the reduction in entropy (uncertainty) about X after observing Y !

Summary

- Probability distributions, both discrete and continuous, play a central role in modeling and perception tasks for robotics
- Bayes rule helps us integrate prior knowledge and accumulate confidence in our estimates of unknown variables
- Information can be measured, based on the characteristics of the distribution from which it is obtained
- These topics come up again and again in machine learning, Bayesian inference, state estimation, communication theory, and most importantly, robotics

But why is it always the Gaussian?

- Lindeberg–Lévy Central Limit Theorem.

Suppose $\{X_1, X_2, \ldots\}$ is a sequence of IID random variables with

$$\mathrm{E}[X_i] = \mu \text{ and } \mathrm{Var}[X_i] = \sigma^2 < \infty.$$

Then as n approaches infinity, the random variables $\sqrt{n}(Sn - \mu)$ converge in distribution to a normal distribution $N(0,\sigma 2)$

